Centralized Rineya Titan 0.6B: laptop execution revision

30 September 2026, America/Chicago. **The inherited 7.4 GiB admission guard was not measured Titan RAM demand. Exact GPU loading and frozen inference have now run successfully on this laptop, with observed process-tree RAM peaks below 2 GiB. Full E0 stopped at a padded-likelihood numerical check. No adapter update, generation, acquisition or retention result has been measured; no recipe is promoted.**

Exact primary student and source

Primary is Rineya-Titan-Experimental-1.0.0-devfreeze: 596,049,920 unique parameters, 310 serialized BF16 tensors, 28 layers, hidden size 1024, MLP 3072, 16 query heads, 8 KV heads, head dimension 128 and vocabulary 151936. Native loading verified the embedding tie and exact copied values. Declared context 40960 is not a qualified operating envelope. The static BF16 payload is 1,192,099,840 bytes, or 1.110 GiB.

Weight SHA256: `537201676814ade103c4efa63f5ebe758ea309971803c4ad633dbca5b6b531b7`. Config SHA256: `660db3b73d788119c04535e48cf9be5f55bc3100841a718637ae695b442f27dd`. Tokenizer SHA256: `aeb13307a71acd8fe81861d94ad54ab689df773318809eed3cbe794b4492dae4`. Template SHA256: `a55ee1b1660128b7098723e0abcd92caa0788061051c62d51cbe87d9cf1974d8`.

The exact local source remains the published Titan devfreeze checkpoint in Chat-Bubble/Rineya_Published_Cores. The dedicated execution branch is research/post-training-mimo-centralized-v1, descended from verified parent `03db218ceebc61a2bc04f2482cf7e094045e01c1`. No parent/main merge or history rewrite occurred. Historical 344M evidence remains preserved and outside primary training. Native runs below used `79ad1a61ba1bb16b93ccc9f0cd27c0e22b9bb00c`. The tested streamed implementation was committed at `4526d705` (abbreviated; full identity is retained in its run receipt and Git). Final documentation and launcher identities are provided by the delivery commit.

Why the RAM policy changed

The user explicitly authorized changing the RAM requirement and plan for a 16 GB laptop, with priority given to VRAM-heavy work. Source audit traced the old 7.4/6 GiB available-memory floors to PRLC project defaults for a GPU model plus frozen CPU reference. The central loader has no full CPU reference: it constructs BF16 weights directly on CUDA and hashes through small chunks. Earlier admission stops therefore established compliance with a policy, not a measured requirement for 7.4 GiB free RAM or a finding that Titan cannot fit.

Rank-8 q/v adapters contain 1,146,880 FP32 parameters: 4.375 MiB. Parameters, gradients and two Adam moments total approximately 17.5 MiB, primarily on GPU. Activation and framework peaks still need measurement. Quantization or CPU optimizer offload has not been added to this first route.

AMENDMENT_009 registered a bounded pilot with 2 GiB system reserve and 2.5 GiB worker RSS cap. Admission reserves all remaining permitted worker growth rather than a fixed 7.4 GiB buffer. After actual loading succeeded, AMENDMENT_010 separated backward/resume memory measurements from the unresolved padded-forward/sampler qualification. The numerical tolerances remain frozen.

AMENDMENT_011 replaces mapped CPU tensor access with a fixed 2 MiB read buffer. It validates the BF16 header, offsets, shapes and complete payload coverage, copies synchronously to GPU and checks every loaded tensor hash. Its named profile keeps 1.25 GiB available system RAM and caps process-tree RSS at 1.75 GiB. Admission is 1.25 + max(0, 1.75 - current RSS) GiB: a 3 GiB before-worker budget. These are prospective measured-trial bounds; the complete streamed GPU path remains unqualified. External 100 ms monitoring, Windows containment, GPU 4/0.5 GiB floors and disk 4 GiB floor remain active. No user application was terminated.

Native results and remaining limits

Attempt	Measured outcome	Peak process RAM	Interpretation
Load pilot	Exact load and finite 256-token forward; 10.406 s worker /11.219 s supervisor	1.928 GiB	Load/inference scope only; CUDA allocation peak 1.400 GiB
Full laptop v1	Exact load and 32 zero-adapter cases, max difference 0; padding check failed at 15.656 s /16.546 s	1.951 GiB	All recorded resource checks passed; no backward/update/generation
Instrumented full v2	Preload RAM 2.515 GiB versus dynamic 3.831 GiB; 7.891 s /8.719 s	0.673 GiB	No model work; numerical error magnitude not remeasured
Streamed load v1	Preload RAM 2.110 GiB versus dynamic 2.331 GiB; 5.078 s /5.875 s	0.673 GiB	New loader not reached; no model work

The full attempt failed its first left-padding likelihood comparison against maximum error 0.03125 and mean error 0.01. Its original receipt did not store the error magnitude. Failure-only telemetry now preserves errors and uses null for nonfinite entries; the instrumented retry stopped before loading. Source inspection found aligned positions and plausible masks. BF16 shape-dependent rounding is a hypothesis, not an established cause. No threshold was relaxed.

Two non-author assistant reviews checked native source/config/journal/cost and scope consistency; this is not human approval. Native verified loads total two, optimizer steps zero, training/generation/teacher tokens or calls zero. The 71 current stdlib controls and nine actual tiny CPU loader controls passed; earlier 12 tiny numerical controls remain a separate run. One historical-helper test initially failed after adding a new dependency; the corrected fixture verifies actual old dependencies and rejects inventing a dependency in an old commit. The failed log/receipt remains preserved. CPU controls do not establish GPU fit.

App-independent execution and detailed sequence

The last OS-only observation found 1.037 GiB available and no Python workers. Further same-state native attempts stopped. A regular PowerShell launcher can run outside the desktop app. It waits at most ten minutes for two observations at least 3.0625 GiB, including supervisor allowance, then invokes one contained 30-minute backward_resume mechanics trial. It aborts on source/config changes, resource failure, timeout or numerical failure and records durable results. Desktop working-set observations near 2 GiB do not guarantee that closing it reclaims exactly that amount. Actual headroom is checked before model work.

The default partial stage retains the exact loader, base and zero-adapter comparisons, positional EOS/PAD label mechanics, 256/512/1024 gradients, five disposable optimizer calls, checkpoint save/reload and bitwise within-stage resume. It omits unresolved padded-forward/causality and sampler checks and explicitly records the changed sampler RNG trajectory. Its distinct resource-only PASS cannot satisfy full E0 or authorize E1. See LAPTOP_RUNBOOK and the launcher files.

After resource and numerical qualification, bind measured laptop successors for native collection, queue and E1. Prioritize one resident GPU worker, pretokenized streamed microbatches, GPU adapters/optimizer state, adapter-only checkpoints and sequential evaluation panels. Avoid retaining a second CPU model or concurrent CPU-heavy data preparation. Register activation checkpointing or shorter contexts only if measured VRAM requires them. Existing E1 plans remain historical until rebound to qualified current source/runtime.

Prepared science and milestone audit

Preparation remains available: 4096 acquisition TRAIN parents, 768 deterministic and 256 curated language replay rows; 5120 total sequences with 680532 nonpadding and 63546 supervised tokens. The historical seed-101 plan has 256 steps, each 16 acquisition plus 4 replay, 13 warmup steps, final LR zero and final-only selection. Later seeds 211/307 require its registered DEV screen.

The DEV prompt/scorer binding contains 832 rows over 544 parent groups and 890874 prompt tokens, with evaluator references separate. Language has 64 DEV/16 CAL candidate worlds and unassigned review packets; grading remains unqualified. Long-context DEV has 32 worlds and 256 related rows at exact 2024/4072 prompt lengths; native 2048/4096 preservation remains unqualified. Parent grouping, source-quality exclusions, original failures and scoped fingerprint checks are retained in their existing receipts and ledgers. CHECK/FINAL remain closed.

Milestone item	Current evidence	State	Remaining work
Current state	Laptop route revised, two exact native loads	Partial progress	Numerical/resource qualification
Exact model	Titan devfreeze; 596049920 parameters	Bound	Same checkpoint retained
Git/source	Native producer 79ad1a61; streamed implementation 4526d705	Committed	Delivery commit records final docs
Verified	Exact load/tie, 32 zero-adapter matches, CPU controls	Scoped	Full E0 not passed
Unmeasured	Backward, optimizer/resume, capability	Open	Bounded next phase
Completed experiment	Exact-load pilot and full attempt through padding	Recorded	Failures retained
Exact result	RAM peak 1.928/1.951 GiB; padding gate failure	Measured	Error diagnosis
Acquisition effect	No native training	Unmeasured	Qualified E1
Retention effect	No scored native preservation	Unmeasured	Capture and grading
Compute/memory	Four workers 39.031 s; supervisors 42.359 s	Measured wall	GPU time/throughput not inferred
Failure modes	Padded likelihood and later headroom drop	Recorded	No tolerance or guard auto-relaxation
Interpretation	7.4 GiB was inherited reserve, not model demand	Source audit	Training fit still needs proof
Ruled out	Admission failures alone prove Titan needs 7.4 GiB free	Invalid inference	No efficacy conclusion
Next experiment	Standalone bounded backward/resume and padding diagnostic	Registered	Actual RAM admission
Heavy compute justified	Bounded qualification	Conditional	E1 waits for scientific gates

The original capability/retention research objective remains incomplete. Current best remains unchanged Titan with null acquisition and retention effects. No preparation or partial resource pass promotes a recipe. Historical blocked audits and 7.4/6 GiB configs remain evidence of earlier conditions; they are superseded operationally by the user's authorized laptop route, not rewritten into successes.